

MERA — Multimodal Emotion Recognition Agent

Enrique Posada

Abstract—Automatic emotion recognition in conversational settings requires the integration of complementary signals from multiple modalities, as no single channel reliably captures the full complexity of human affective expression. This work presents a multimodal emotion recognition system combining a CNN-based vision expert trained on FER2013+, a RoBERTa-BiLSTM text expert trained on GoEmotions, and two independent fusion strategies evaluated on the MELD benchmark. The first fusion approach employs a GRU-based manager network trained on pre-extracted expert logits to model temporal visual context across utterance frames. The second integrates both expert predictions with a knowledge-based system implemented in SWI-Prolog, which applies explicit psychological reasoning rules to detect phenomena including emotional masking, irony, and neutral override — cases where statistical fusion approaches do not model inter-modality relationships explicitly. Hyperparameters for all learned components are determined through Accelerated Particle Swarm Optimization. The vision expert achieves a macro F1 of 0.75 on FER2013+ and the text expert achieves 0.58 on GoEmotions, with both showing expected performance degradation upon transfer to MELD’s in-the-wild conversational domain. The knowledge-based system is evaluated qualitatively, demonstrating emotionally appropriate conversational responses driven by a locally deployed large language model.

Index Terms—Multimodal Emotion Recognition, Knowledge-Based System, Particle Swarm Optimization.

I. INTRODUCTION

EMOTION plays a fundamental role in human communication. The ability to automatically recognize emotional states has become increasingly relevant across applications ranging from human-computer interaction and mental health monitoring to adaptive intelligent systems, and has seen remarkable progress driven by advances in deep learning and multimodal data processing [1].

Human emotions are inherently multimodal — expressed simultaneously through facial expressions, language, and vocal tone. Systems relying on a single modality are fundamentally limited: a speaker may choose neutral words while their face conveys distress, or text alone may be ambiguous without visual context [2]. This has motivated multimodal emotion recognition systems that combine complementary signals for more robust predictions.

Fusion architecture is a central design choice in such systems. Early fusion integrates raw features before classification, capturing cross-modal correlations but requiring strict temporal alignment. Late fusion combines outputs of independent modality-specific models, preserving modularity and robustness to missing modalities [1]. A further gap exists between recognition research and conversational deployment — most dialogue agents respond to semantic content while remaining blind to the user’s emotional state, making interactions feel impersonal.

In this work, we address these challenges through two independent fusion strategies built on decoupled vision and text experts. The first is a GRU-based manager network that processes temporal sequences of vision logits and anchors them with text at message-send time. The second is a knowledge-based system implemented in SWI-Prolog that applies explicit psychological reasoning rules to detect inter-modality phenomena such as emotional masking and irony. Both are integrated with a locally deployed large language model for emotion-aware conversational interaction.

The main contributions are threefold. First, a late-fusion manager network with GRU-based temporal modeling and optional text anchoring for missing modality robustness. Second, a knowledge-based system that explicitly models inter-modality conflict cases that purely statistical approaches do not capture. Third, end-to-end integration with a local language model enabling emotion-aware dialogue without external data transmission.

The remainder of this paper is organized as follows. Section II reviews the State of the Art regarding emotion recognition across single and multiple modalities. Section III details our Proposal and the approach taken during the project. Section IV presents the Evaluation and results, and finally, Section V provides our Conclusions and Section VI the directions for Future Work.

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II. STATE OF THE ART

A. Emotion Representation (ER)

Establishing a formal framework for representing emotional states is a prerequisite for any automatic emotion recognition system. Two dominant paradigms have shaped the field: categorical and dimensional representations.

On one hand, the categorical or discrete approach, most notably formalized by Ekman’s Basic Emotion Theory, postulates a small set of biologically grounded, universally recognizable basic emotions typically anger, disgust, fear, happiness, sadness, and surprise. Argued to have evolved to solve fundamental adaptive problems and expressed through distinctive patterns in facial musculature and physiological responses [3]. This discrete schema offers clear advantages for computational systems: interpretability, straightforward classifier design, and direct alignment with the annotation conventions of most benchmark datasets [4]. Its primary limitation is coarse granularity — mixed or ambiguous emotional states are difficult to represent, and the model is sensitive to cultural variation.

On the other hand, dimensional approach is represented by Russell’s circumplex model, instead conceptualizes af-

fect as points in a continuous low-dimensional space defined by valence (pleasure–displeasure) and arousal (activation–deactivation), with some authors adding a third dominance axis [5]. This representation captures emotional intensity, blends, and temporal dynamics that discrete labels miss. Plutchik’s psychoevolutionary wheel extends this view by arranging eight primary emotions in opponent pairs on a color wheel, where radial distance encodes intensity and adjacent combinations produce blended states such as optimism or remorse [6]. These models are particularly prevalent in physiological signal-based recognition and continuous affect tracking applications.

This work adopts the categorical paradigm with Ekman’s seven basic emotions, adding neutral to the original six, as this schema is shared across all three datasets used in this work and provides the discrete label space required by the knowledge-based fusion system. The choice of macro F1 as the primary evaluation metric, discussed in Section III, reflects a direct consequence of this categorical framing — treating all seven classes as equally important regardless of their frequency in the training distribution.

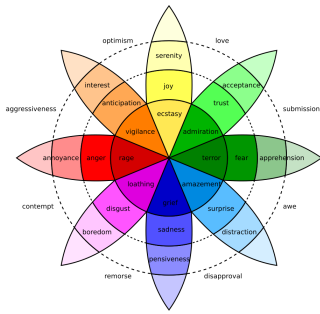


Fig. 1. Plutchik’s psychoevolutionary wheel

B. Facial Emotion Recognition

Facial emotion recognition (FER) is the most obvious method of ER and consists in classifying someone’s emotion based on his facial expression. The human face is a highly expressive medium for nonverbal communication, capable of showing emotional shifts through muscular movements, gaze direction, and micro-expressions [1].

Early FER approaches relied on handcrafted descriptors to capture expression-relevant features from facial images. The Histogram of Oriented Gradients (HOG) encodes edge orientations and shape variations caused by facial muscle movements, while Local Binary Patterns (LBP) capture local texture patterns by thresholding pixel intensities relative to their neighbours [7][8]. Variants such as LBP-TOP extended these descriptors to the spatiotemporal domain to capture dynamic expressions in video sequences. These methods are computationally efficient and interpretable but degrade significantly under illumination changes, head pose variation, and occlusion — conditions that are ubiquitous in real-world footage.

A parallel line of work introduced the Facial Action Coding System (FACS), developed by Ekman and Friesen, which

describes facial expressions in terms of discrete muscle activations called Action Units (AUs) [1]. Emotions are represented as specific AU combinations. AUs serve as interpretable mid-level features bridging raw pixel data and categorical emotion labels, and are robust to identity differences since they describe motion patterns rather than static geometry [9].

Deep learning approaches have largely improved hand-crafted methods by learning hierarchical discriminative representations directly from raw images. 2D-CNNs such as VGGNet and ResNet extract spatial features from static facial frames, while 3D-CNNs extend convolutions into the temporal dimension to capture spatiotemporal motion patterns associated with dynamic expressions [10], [11]. More recently, Vision Transformers (ViTs) have emerged as a powerful alternative, applying self-attention mechanisms to sequences of facial image patches and capturing long-range spatial dependencies across facial regions that convolutional architectures may miss [12]. Pretraining on large-scale datasets such as AffectNet and RAF-DB followed by task-specific finetuning has proven effective in data-scarce scenarios [13].

A persistent challenge across all approaches is robustness to head pose and illumination variation. Changes in head orientation partially occlude key facial regions such as the eyes and mouth, while inconsistent lighting conceals subtle texture patterns and micro-expressions. Pose normalisation techniques — including 3D morphable models and GAN-based frontalisation — and illumination-invariant preprocessing have been explored to mitigate these effects, with recent work combining them with ViT and 3D-CNN architectures showing improved performance on in-the-wild benchmarks [1].

In this work, the vision expert follows the lightweight CNN paradigm — a custom VGG-style network trained on FER2013+ that processes each video frame independently, producing per-frame emotion logits.

C. Text-based Emotion Recognition

Text-based emotion recognition (TER) identifies emotional states from written or spoken language, relying exclusively on linguistic attributes such as lexicon, syntax, and semantic context rather than visual cues [1]. TER applications span opinion mining and market analysis, affective computing, and human-computer interaction [14]. This work focuses on the latter — developing an emotion-aware conversational agent that adapts its responses to the user’s detected affective state in real time.

Early approaches relied on lexicon-based methods, where curated affective dictionaries assign emotional categories or continuous valence-arousal-dominance scores to individual words. Notable resources include the NRC Emotion Lexicon [15], WordNet-Affect [16], and the Affective Norms for English Words (ANEW) [17]. Rule-based extensions introduced negation handling, intensifier recognition, and domain-specific patterns to improve coverage [18]. While interpretable and computationally lightweight, these methods are fundamentally limited by their context insensitivity — the same word carries different emotional weight depending on surrounding discourse — and their inability to detect sarcasm, where literal

word meaning directly contradicts the intended emotional tone [1].

Classical machine learning approaches operating on bag-of-words, TF-IDF vectors, and n-gram features with Support Vector Machines and Naïve Bayes classifiers improved upon lexicon-based methods but remained unable to capture long-range dependencies and contextual meaning [19], [20]. Deep learning architectures addressed these limitations, with CNNs and LSTM networks capturing local and sequential dependencies respectively [21]. The introduction of Transformer-based models — notably BERT [22], RoBERTa [23], and GPT-based architectures [24] — established new performance standards by leveraging bidirectional self-attention to model long-range contextual interactions across entire sequences. Pretraining on large corpora followed by finetuning on emotion-labelled datasets such as GoEmotions enables these models to adapt to specific recognition tasks with limited labelled data [25], [26].

A persistent challenge across all TER approaches is the accurate interpretation of sarcasm and context-dependent meaning. Sarcastic expressions may convey negative affect despite positive surface vocabulary, requiring reasoning beyond lexical content. Multimodal systems that combine text with visual or prosodic cues have shown improved sarcasm detection [1] — a direct motivation for the multimodal fusion approach adopted in this work, where the knowledge-based system explicitly models the discrepancy between facial affect and linguistic content as an irony signal.

In this work, the text expert adopts the RoBERTa-BiLSTM architecture — a RoBERTa encoder providing deep contextual embeddings followed by a BiLSTM head for sequence-level aggregation — trained on the GoEmotions dataset mapped to the seven-class Ekman schema. RoBERTa was selected over BERT for its more robust pretraining procedure, which removes the next-sentence prediction objective and uses dynamic masking [23].

D. More Emotion Recognition Modalities

Beyond facial expressions and text, affective computing research has explored additional modalities for emotion recognition. Speech-based recognition leverages prosodic and acoustic features — including pitch, energy, and MFCCs — to infer emotional states independently of linguistic content [1], with deep learning approaches extending toward spectrogram-based CNN and Transformer architectures [27], [28]. However, a well-documented limitation is sensitivity to cross-corpus generalization [1] — models trained on acted datasets such as RAVDESS degrade significantly on in-the-wild data such as MELD, a limitation directly encountered in this work where a spectrogram-based speech expert was developed but discarded due to poor performance. Physiological signals such as EEG, GSR, ECG, and EMG [1] are particularly resistant to conscious manipulation, making them valuable for high-stakes applications, but require specialized wearable sensors that severely restrict deployment in naturalistic HCI scenarios. For this reason, this work targets a fully non-contact, camera and microphone based paradigm.

E. Multimodal Emotion Recognition and Fusion Strategies

The limitations of unimodal approaches — occlusion and pose sensitivity in facial recognition, sarcasm and contextual ambiguity in text, among others — have motivated a substantial body of research into multimodal emotion recognition (MER), where two or more modalities are combined to produce more robust and reliable affective estimates than any single modality can achieve alone [1]. The central challenge in MER is not the individual recognition of each modality but the design of the fusion mechanism that combines their outputs effectively.

Early, late, and hybrid fusion. Three broad fusion paradigms dominate the literature. Early fusion, also known as feature-level fusion, concatenates raw or lightly processed features from multiple modalities into a single joint representation before classification [1]. This approach allows the model to learn cross-modal correlations at the feature level but requires modalities to be temporally aligned and is sensitive to missing or noisy inputs — if one modality is absent at inference time, the entire representation is corrupted. Late fusion, also known as decision-level fusion, trains independent unimodal models and combines their predictions or output probability distributions through learned or fixed aggregation rules [1]. This design is more robust to missing modalities since each expert can operate independently, and it allows modality-specific architectures to be optimised separately. Hybrid fusion approaches combine elements of both, typically extracting unimodal features independently before fusing them at an intermediate representation layer [1]. While hybrid methods can capture both low-level cross-modal correlations and high-level semantic agreement, they introduce greater architectural complexity and training difficulty.

Transformer-based fusion. The success of attention mechanisms in unimodal settings has been extended to multimodal fusion through cross-modal Transformer architectures. The Multimodal Transformer [29] introduced cross-modal attention that directly attends across modality sequences, allowing each modality to be conditioned on the temporal dynamics of the others without requiring explicit alignment. Subsequent work has explored factorized attention, modality-specific positional encodings, and hierarchical fusion schemes that operate at multiple temporal scales [1]. These approaches have achieved strong results on benchmarks such as CMU-MOSI and CMU-MOSEI but typically require all modalities to be present and aligned during both training and inference.

Temporal modeling in conversational emotion recognition. Emotion in conversation is not a static property of isolated utterances — it evolves dynamically across turns, influenced by the emotional history of the interaction, the relationship between speakers, and the broader conversational context [2]. This observation motivated recurrent architectures specifically designed for emotion recognition in conversations (ERC). DialogueRNN [30] models the emotional state of each speaker as a hidden state updated by a GRU at each turn, conditioned on the global conversational context and the emotional states of other participants.

Missing modality robustness. A practical challenge in MER

systems is that not all modalities are reliably available at inference time. Occlusion, microphone failure, or the absence of spoken content can render individual modality streams unavailable or unreliable. Several approaches have addressed this through modality dropout during training [1]. This robustness consideration directly influenced the architecture of this work, where the text modality is treated as an optional anchor — the system operates on vision alone when no text input is available and incorporates text only at the moment of user interaction.

Knowledge-based and rule-based fusion. While the majority of MER research focuses on learned fusion mechanisms, a complementary line of work draws on symbolic reasoning to combine modality predictions. Knowledge-based systems encode domain expertise about inter-modality relationships as explicit rules, providing interpretable and deterministic fusion decisions that do not require labelled training data for the fusion component itself. This approach is particularly well-suited to capturing psychological phenomena such as emotional masking — where facial affect contradicts verbal content — and irony detection, where the combination of a negative facial expression with positive language constitutes a meaningful signal that purely statistical fusion methods may not reliably capture. The interpretability of rule-based fusion is a further advantage in human-computer interaction applications, where understanding why the system made a particular decision is as important as the decision itself.

In this work, two independent fusion approaches are implemented. The first is a late fusion architecture based on a manager network — a GRU-based model trained on pre-extracted expert logits from the MELD dataset that learns to combine temporal sequences of vision predictions with utterance-level text predictions. The second fusion method employs both expert predictions alongside a knowledge-based system, which applies explicit psychological reasoning rules to derive a dominant emotion and response strategy — capturing phenomena such as emotional masking, irony, and neutral override that lie beyond the scope of purely statistical approaches.

F. Datasets for Emotion Recognition

The availability of annotated corpora is a fundamental prerequisite for training and evaluating emotion recognition systems. Datasets vary substantially in their collection conditions, modalities, annotation schemas, and emotional distributions, making benchmark selection a critical decision that directly affects the comparability of results across works [1].

FER2013 and FER2013+. FER2013 [31] is one of the most widely adopted benchmarks for facial expression recognition, consisting of 35,887 grayscale face images at 48×48 pixel resolution collected through automated web searches and manually labelled with seven emotion categories. Despite its scale, FER2013 is known for its noisy single-annotator labels, which introduce significant label uncertainty particularly for ambiguous or subtle expressions. FER2013+ [32] addresses this limitation by reannotating each FER2013 image through crowd-sourced labelling across ten independent annotators, recording the full vote distribution across nine categories

per image rather than assigning a single label. This cleaner annotation scheme makes FER2013+ the preferred choice for training facial expression models, and it is the dataset used to train the vision expert in this work.



Fig. 2. FER2013 Images sample

GoEmotions. GoEmotions [33] is a large-scale text emotion dataset consisting of 58,009 English Reddit comments annotated with 28 fine-grained emotion categories by trained raters. GoEmotions is notable for its scale, its diversity of informal language styles, and its coverage of subtle and mixed emotional states that coarser datasets do not capture. Its Reddit origin introduces domain-specific linguistic patterns — informal register, sarcasm, and internet idioms.

RAVDESS. The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) [34] contains 7,356 audio-visual recordings of 24 professional actors producing emotionally expressive speech and song across eight emotion categories. Its controlled studio recording conditions yield high-quality, unambiguous emotional performances, making it a popular benchmark for speech and audio-visual emotion recognition. However, the acted and controlled nature of RAVDESS introduces a pronounced domain gap when models are applied to in-the-wild conversational datasets — a limitation directly encountered in this work, where a speech expert trained on RAVDESS failed to generalize to MELD and was subsequently discarded.

IEMOCAP. The Interactive Emotional Dyadic Motion Capture database (IEMOCAP) [35] is a multimodal dataset containing approximately 12 hours of audiovisual recordings of dyadic interactions between ten actors, annotated with categorical emotion labels. Its scripted and improvised conversation scenarios and multi-annotator labelling make it one of the most rigorous benchmarks in conversational emotion recognition. IEMOCAP is widely used as a standard evaluation benchmark in the MER literature providing a reference point for comparing system performance across works, though it is not used directly in this work.

MELD. The Multimodal EmotionLines Dataset (MELD) [36] was constructed from the TV series *Friends* and contains 13,708 utterances from 1,433 dialogues, annotated with seven emotion categories aligned with the Ekman schema. MELD is one of the few large-scale conversational emotion datasets providing aligned video, audio, and text modalities in a naturalistic multi-party dialogue setting, making it particularly suitable for evaluating multimodal fusion systems in realistic conditions. Its primary challenges include severe class imbalance and the domain characteristics of TV series production, including variable camera angles, background music, audience noise, and overlapping speakers. MELD serves in this work as the finetuning target for both unimodal experts, the training dataset for the manager network, and the primary evaluation benchmark for the manager network.

G. *Emotion-Aware Conversational Systems*

The integration of affective awareness into conversational agents represents a natural extension of emotion recognition research toward practical human-computer interaction applications. While early dialogue systems operated purely on semantic content — responding to what the user said without any consideration of how they were feeling — a growing body of work has demonstrated that incorporating emotional context substantially improves user satisfaction, engagement, and perceived naturalness [18].

Early emotion-aware dialogue systems relied on rule-based sentiment detection combined with hand-crafted response templates, selecting responses from a predefined pool based on detected sentiment polarity [18]. While straightforward to implement and interpret, these systems were severely limited by their inability to generate novel responses and their coarse sentiment granularity — distinguishing only between positive and negative affect rather than the full spectrum of discrete emotions. The advent of neural sequence-to-sequence models enabled end-to-end trainable dialogue systems that could condition response generation on emotional context, with works such as MoEL introducing emotion-conditioned decoders that explicitly control the emotional tone of generated responses [37], [38].

The emergence of large pretrained language models — BERT, GPT, and their successors — shifted the paradigm toward retrieval-augmented and generative approaches that leverage vast pretraining knowledge for more natural and contextually appropriate responses. More recently, large language models such as LLaMA and GPT-4 have demonstrated remarkable zero-shot and few-shot capability for emotionally nuanced dialogue, responding to emotional cues implicitly encoded in the conversation history without explicit emotion labels. However, these models rely on the linguistic content of the conversation alone — they have no access to the user’s facial expression or vocal affect, limiting their ability to detect masked emotions, irony, or discrepancies between what the user says and how they actually feel.

Multimodal conversational systems address this limitation by incorporating non-linguistic signals into the response generation pipeline. A common architecture involves a separate emotion recognition module whose output is injected into the language model as a soft prompt or system instruction, decoupling the recognition and generation components and allowing each to be developed and evaluated independently. This decoupled design is particularly advantageous when the emotion recognition module is itself multimodal — the language model does not need to process raw audio or video, only the symbolic output of the recognition pipeline.

A further distinction exists between systems that reason about emotion internally within the language model and systems where emotional reasoning is performed externally by a dedicated module before response generation. The former approach is more flexible but less interpretable and harder to control — the model may ignore or misinterpret emotional signals depending on how they are presented in the prompt. In this work, a locally deployed Ollama language model serves

as the conversational backbone, receiving emotional context derived from the knowledge-based fusion system as part of its system prompt at each interaction turn. This architecture deliberately avoids reliance on cloud-based APIs, ensuring user privacy and enabling fully offline deployment — a practical consideration for sensitive affective computing applications where conversation content may be personal or distressing.

H. *Hyperparameter Optimization with Metaheuristics*

The performance of deep neural networks is highly sensitive to hyperparameter choices — learning rate, dropout rate, hidden layer dimensions, and architectural decisions such as the number of recurrent layers all significantly affect convergence behaviour and generalization. Manual tuning of these parameters through trial and error is impractical given the exponential size of the combined search space, motivating the use of automated optimization strategies [39].

Metaheuristic algorithms offer a family of optimization strategies inspired by natural phenomena — including evolutionary processes, swarm behaviour, and physical annealing — that are well-suited to non-convex, high-dimensional search spaces without requiring gradient information or differentiability of the objective [39]. These algorithms operate by maintaining a population of candidate solutions that evolve iteratively according to update rules designed to balance exploration of the search space with exploitation of promising regions. Several metaheuristic families have been applied to neural network hyperparameter optimization, including Genetic Algorithms, Ant Lion Optimization, Artificial Bee Colony, and Bat Algorithm, with comparative studies showing consistent improvements over manual tuning across a range of CNN architectures and datasets [40].

Particle Swarm Optimization (PSO) [41] is among the most widely adopted metaheuristic algorithms for hyperparameter tuning. Inspired by the collective movement of bird flocks and fish schools, PSO maintains a swarm of particles where each particle represents a candidate hyperparameter configuration. Particles move through the search space according to their own best-known position and the best-known position of the entire swarm, iteratively converging toward optimal regions. PSO has demonstrated strong performance on CNN hyperparameter optimization tasks due to its fast convergence and low number of algorithm-level parameters [40].

Accelerated Particle Swarm Optimization (APSO) extends standard PSO by simplifying the velocity update equation and introducing a global best attraction mechanism that accelerates convergence while maintaining population diversity. APSO has been applied to CNN hyperparameter optimization showing competitive results with fewer iterations than standard PSO [42]. In this work, APSO is adopted as the hyperparameter optimization strategy for all three learned components of the system — the vision expert, the text expert, and the manager network — with each expert optimized independently in a decoupled fashion.

III. APPROACH

A. System Overview

This work proposes a multimodal emotion recognition system combining a facial expression expert and a text expert. The work explores the late fusion of both modalities through two independent approaches — a manager network that learns to combine expert logits through a GRU-based temporal module, and a knowledge-based system that applies explicit psychological reasoning rules over expert predictions to derive a dominant emotion and response strategy — with the KBS integrated into an emotion-aware conversational agent powered by a locally deployed large language model. The system is built around three core principles: modality decoupling, temporal emotional context, and robustness to missing modalities.

The architecture consists of five main components: a face detection and preprocessing pipeline, a vision expert (VisionNet), a text expert (RoBERTa-BiLSTM), a late-fusion manager network (ManagerNet), and a knowledge-based system (KBS) connected to a locally deployed language model via Ollama. The manager network was initially intended as the conversational backbone, but its limited performance on MELD motivated the integration of the KBS as an independent fusion strategy.

During development, an audio expert based on mel spectrogram features was also explored as a third modality. However, as discussed in Section III-E, it was discarded due to poor performance. The final system operates on vision and text modalities only.

B. Training Datasets and Class Imbalance Mitigation

Each expert is trained on a dedicated dataset chosen for its alignment with the expert’s modality and label schema. A consistent set of seven emotion categories is used throughout the system — neutral, happiness, surprise, sadness, anger, disgust, and fear — corresponding to Ekman’s basic emotion taxonomy.

Table I reports the class distribution across the three training splits. All three datasets exhibit class imbalance, with neutral dominating MELD at 47.15% while fear and disgust together represent under 6%. GoEmotions shows the strongest happiness bias at 39.04%. These imbalances directly motivated the use of WeightedRandomSampler and macro F1 as the primary optimization metric throughout all training pipelines.

Vision expert — FER2013+. The vision expert is trained on FER2013+, an improved version of the original FER2013 dataset that replaces single-annotator labels with majority voting across ten crowd-sourced annotators per image. A winner-takes-all labeling strategy is applied: for each image, the most-voted emotion is selected as the label, and samples where the plurality vote falls on contempt, unknown, or not-a-face categories are discarded. This produces a cleaner label distribution than the original FER2013 while preserving the 48×48 grayscale image format. Data augmentation during training includes random horizontal flipping, small rotations ($\pm 10^\circ$), brightness and contrast jitter, and random affine translations and scaling, all chosen to simulate realistic variations present in in-the-wild video footage.

Text expert — GoEmotions. The text expert is trained on GoEmotions, a large-scale corpus of Reddit comments annotated with 28 fine-grained emotion categories. These are mapped to the 7-class Ekman schema using a provided mapping file that groups fine-grained emotions into their corresponding basic emotion categories. To address class imbalance in the mapped distribution, asymmetric random word swap augmentation is applied during training: utterances belonging to minority classes disgust and fear are augmented with a swap probability of 0.25, while standard classes use a probability of 0.10. This provides additional lexical diversity for underrepresented categories without distorting the semantic content of majority class samples. A WeightedRandomSampler is applied in both expert training pipelines to ensure minority classes appear with equal expected frequency per epoch.

Initial vs final training setup. Initial training experiments for both experts used standard cross-entropy loss, uniform sampling, minimal augmentation, and accuracy as the optimization metric. This produced models heavily biased toward majority classes, with poor F1 scores on fear, disgust, and surprise. The final training setup addressed these issues through three changes: WeightedRandomSampler to balance class exposure per epoch, extended data augmentation as described above, and macro F1 as the optimization metric, which treats all classes equally regardless of frequency. These changes produced measurable improvements in minority class coverage across both experts.

Manager and finetuning dataset — MELD. The Multimodal EmotionLines Dataset (MELD) is used as the benchmark for evaluating the manager system and as the finetuning target for both experts. MELD was constructed from the TV series Friends and contains 11,098 utterances across training, development, and test splits, annotated with the same seven emotion categories used throughout this work. Unlike FER2013+ and GoEmotions, which were collected under relatively controlled conditions, MELD presents significant challenges for both modalities. The video tracks inherit the characteristics of TV series production — variable camera angles, partial occlusions, background noise, compression artifacts, and multiple simultaneous speakers — representing a substantial domain shift from the controlled facial expression images in FER2013+. Furthermore, MELD exhibits severe class imbalance: neutral utterances account for approximately 47% of the training split, while fear and disgust together represent under 6%, as shown in Table I. This imbalance directly limits the performance of all trained models on minority classes and is a known limitation acknowledged in the multimodal emotion recognition literature. Both experts were finetuned on MELD’s training split using the WeightedRandomSampler and macro F1 optimization strategy described above, with the development split used for validation. The test split was withheld exclusively for final evaluation. The original MELD test split was corrupted, so a replacement was constructed by sampling 1,000 utterances from the training set while preserving class proportions. These samples were excluded from all training and finetuning procedures, ensuring no overlap between the evaluation set and the data seen by any component of the system.

TABLE I
CLASS DISTRIBUTION ACROSS THE THREE DATASETS USED IN THIS WORK. COUNTS AND PROPORTIONS SHOWN FOR TRAINING SPLITS ONLY.

Emotion	FER2013+		GoEmotions		MELD	
	Count	%	Count	%	Count	%
Neutral	10,640	37.06	12,823	29.54	4,710	47.15
Happiness	7,547	26.29	16,948	39.04	1,743	17.45
Surprise	3,593	12.52	4,489	10.34	1,205	12.06
Sadness	3,566	12.42	2,619	6.03	683	6.84
Anger	2,495	8.69	5,336	12.29	1,109	11.10
Fear	658	2.29	615	1.42	268	2.68
Disgust	210	0.73	580	1.34	271	2.71
Total	28,709	100.0	43,410	100.0	9,989	100.0

C. Vision Expert — VisionNet

The vision expert is a custom convolutional neural network designed for per-frame facial emotion classification from grayscale 48×48 images. It follows a VGG-style progressive architecture with three convolutional blocks and a fully connected classification head.

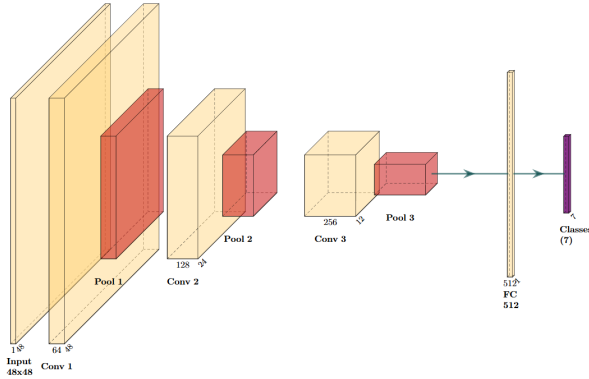


Fig. 3. Vision Expert Architecture

Block 1 consists of two convolutional layers (1→64→64 channels), each followed by batch normalization and ReLU activation, and a 2×2 max pooling layer reducing spatial dimensions from 48×48 to 24×24. Block 2 mirrors this structure with doubled channel depth (64→128→128), reducing to 12×12. Block 3 applies a single convolutional layer (128→256 channels) with batch normalization and max pooling, producing a feature map of shape 256×6×6. The classification head flattens this representation and passes it through a linear projection layer of APSO-optimized size, batch normalization, ReLU, dropout, and a final linear layer producing 7-class logits.

VisionNet is intentionally stateless — it processes each frame independently with no temporal awareness of its own. Temporal reasoning over the sequence of per-frame predictions is delegated entirely to the manager network, keeping the vision expert modular and independently reusable.

The network was trained on FER2013+ using Adam optimizer with the APSO-optimized learning rate and plain cross-entropy loss, for up to 100 epochs with early stopping patience of 20 epochs. A `ReduceLROnPlateau` scheduler with patience of 8 and reduction factor of 0.4 was applied on validation macro F1 to decay the learning rate when improvement

stalled. The best checkpoint was selected by validation macro F1. The network achieved 0.75 macro F1 on the FER2013+ test split. It was subsequently finetuned on MELD training frames using the imbalance mitigation strategies described in the previous subsection, with all network parameters updated jointly at a fixed learning rate of 1×10^{-5} to avoid large weight updates that could destabilize the pretrained representations.

Figure 4 illustrates the training dynamics of VisionNet on FER2013+. The loss curve shows smooth monotonic convergence, while validation macro F1 stabilises around epoch 30 and remains consistent with the final test macro F1 of 75%, indicating healthy generalisation without significant overfitting.

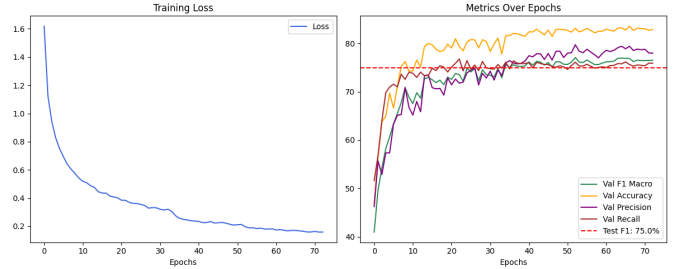


Fig. 4. VisionNet training curves on FER2013+. Left: training loss over epochs. Right: validation metrics over epochs with test macro F1 shown as a dashed reference line (75.0%).

When training the manager or running inference, VisionNet processes all T detected frames of an utterance in a single batched forward pass, producing a sequence of logits of shape $[T, 7]$ passed to the manager network.

D. Text Expert — RoBERTa-BiLSTM

The text expert combines a pretrained RoBERTa-base encoder with a Bidirectional LSTM classification head. Each utterance is tokenized using the RoBERTa-base tokenizer with a maximum sequence length of 64 tokens, with padding and truncation applied as needed.

The RoBERTa encoder produces contextual token embeddings which are fed into a BiLSTM. The hidden states of the BiLSTM are aggregated into a single utterance-level vector using one of three pooling strategies: mean pooling over non-padding tokens (masked average), max pooling with large negative masking of padding positions, or last-state pooling via concatenation of the final forward and backward hidden states. The pooling strategy was treated as a discrete hyperparameter and optimized by APSO alongside the other parameters.

Training employed a differential learning rate strategy via AdamW with weight decay of 0.01 — the BiLSTM head and classification layers were updated at the APSO-optimized learning rate, while the last two RoBERTa encoder layers and the pooler were updated at the lower APSO-optimized RoBERTa learning rate. For the first five epochs all remaining RoBERTa layers were frozen while the BiLSTM head and the last two encoder layers trained freely, allowing task-specific adaptation before the encoder was fully re-frozen for the remaining epochs to prevent overfitting or catastrophic forgetting. A ReduceLROnPlateau scheduler with patience of 3 and reduction factor of 0.5 was applied on validation macro F1, with early stopping triggered after 5 epochs without improvement. Mixed precision training using `torch.amp` with `float16` was employed to reduce GPU memory consumption and thermal load during training, which was necessary given the memory and thermal constraints of the available hardware. The model was trained for up to 15 epochs with a batch size of 64.

Figure 5 illustrates the training dynamics of the RoBERTa-BiLSTM text expert on GoEmotions. The loss curve shows smooth convergence over 13 epochs before early stopping triggered. The visible dip in validation metrics around epoch 5 corresponds to the RoBERTa encoder re-freezing after the initial partial fine-tuning phase, after which metrics recover and stabilise close to the test macro F1 of 57.2%.

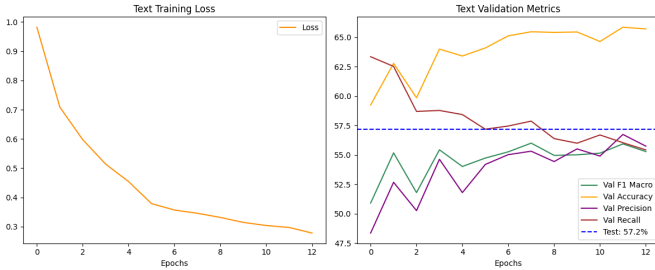


Fig. 5. RoBERTa-BiLSTM training curves on GoEmotions. Left: training loss over epochs. Right: validation metrics with test macro F1 shown as a dashed reference line (57.2%).

The text expert produces a single logit vector of shape [7] per utterance, representing the emotion distribution over the full utterance text. Unlike the vision expert which produces a temporal sequence of T vectors, the text expert produces a single prediction — this asymmetry between output formats is handled by the manager network’s architecture, which treats vision and text through dedicated encoding branches.

E. Audio Expert — Attempted and Discarded

An audio expert was developed as a third modality during the exploration phase of this work. The expert used mel spectrogram features extracted from speech signals at 22050 Hz with 128 mel frequency bins, normalized and padded to a fixed temporal width of 172 frames. The model was trained on RAVDESS using a speaker-independent split, with SpecAugment time and frequency masking as data augmentation and Gaussian noise injection at 20dB SNR for additional robustness.

However, it was deprecated as it exhibited poor convergence on the primary evaluation metrics. Preliminary training runs resulted in a persistent F1-score that remained significantly below the established viability threshold, as shown in Figure 6. Consequently, further development was discontinued to prioritize more promising architectures.

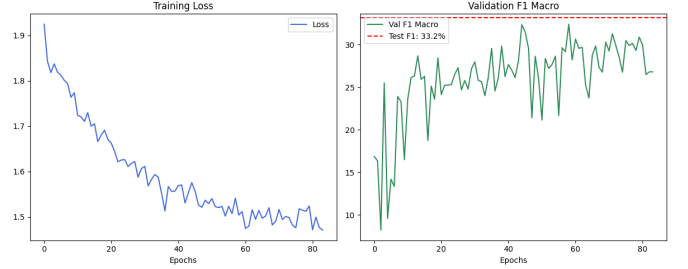


Fig. 6. AudioNet training curves on RAVDESS+. Left: training loss over epochs. Right: validation metrics over epochs with test macro F1 shown as a dashed reference line (33.2%).

F. Experts Finetuning on MELD

Both experts were evaluated on MELD prior to finetuning, revealing a significant performance drop attributable to the domain shift between their respective training datasets and MELD’s in-the-wild TV series footage. The effect was most pronounced in the vision expert, whose macro F1 collapsed from 0.75 on FER2013+ to 0.15 on MELD — close to random guessing for a seven-class problem — as the model defaulted to predicting neutral and happiness for virtually all inputs. The text expert showed a less severe but still substantial drop, achieving a macro F1 of 0.30 on MELD despite never having been exposed to conversational data during training. These results motivated finetuning both experts on MELD to reduce the domain gap before training the manager network, whose performance depends directly on the quality of the input logits it receives.

During vision expert finetuning, all VisionNet parameters were updated jointly using Adam with a fixed learning rate of 1×10^{-5} , chosen deliberately low to avoid large weight updates that could destabilize the FER2013+ pretrained representations. A WeightedRandomSampler with inverse class frequency weights was applied to counteract MELD’s severe class imbalance, combined with the same data augmentation pipeline used during FER2013+ training — random horizontal flipping, rotation (± 10), affine translations, and brightness and contrast jitter. Plain cross-entropy loss was used over 5 epochs with a batch size of 256, with the best checkpoint selected by validation macro F1. Finetuning produced a marginal macro F1 improvement from 0.15 to 0.17 on the MELD test split, driven by gains in previously zero-performance classes such as surprise and fear, at the cost of reduced neutral accuracy — the expected tradeoff of the WeightedRandomSampler redistributing predictive capacity toward minority classes. The finetuned vision expert is used in all subsequent experiments as it better reflects the target domain despite the modest overall improvement.

During text expert finetuning, the RoBERTa encoder was fully frozen and only the BiLSTM head was updated, as

identified by checking which parameter names contained the string `roberta` and setting `requires_grad = False` accordingly. This freezing strategy was selected after experimentation — allowing the full model to finetune led to unstable training and faster degradation of minority class performance, while freezing RoBERTa entirely and updating only the BiLSTM head provided more stable gradient flow. A `WeightedRandomSampler` and plain cross-entropy loss were used over 5 epochs with a learning rate of 2×10^{-5} . However, finetuning was observed to degrade overall macro F1 from 0.30 to 0.28 compared to the original pretrained model on MELD, likely due to the BiLSTM head overfitting to the resampled minority class distribution given its limited parameter count relative to the complexity of MELD’s conversational domain. The original pretrained text expert was therefore retained for all subsequent experiments.

G. Manager Network — *ManagerNet*

The manager network is where both experts meet. It receives the frame-level vision logits $[T, 7]$ from VisionNet and optionally the utterance-level text logits $[7]$ from the text expert, and produces a single fused emotion prediction integrating temporal visual context with linguistic content.

Vision temporal encoder. The sequence of vision logits $[T, 7]$ is first projected to a hidden-dimensional space via a linear layer followed by batch normalization and ReLU, producing a projected sequence $[T, \text{hidden-dim}]$. This sequence is processed by a multi-layer GRU which reads frames in temporal order, accumulating emotional context across the utterance. The final hidden state of the top GRU layer $[\text{hidden-dim}]$ represents the temporal emotional baseline derived from the full sequence of observed facial expressions.

Text anchor branch. When text logits are available, they are projected into the same hidden space via a linear layer, batch normalization, and ReLU, producing a vector $[\text{hidden-dim}]$. When text is unavailable — during live streaming inference before the user sends a message — this branch outputs a zero vector of identical shape. The classifier input dimension is always $[\text{hidden-dim} \times 2]$ regardless of text availability, ensuring architectural consistency across all inference modes without requiring separate model variants.

Fusion and classification. The GRU final state and the text projection (or zero vector) are concatenated into a fused representation $[\text{hidden-dim} \times 2]$, passed through dropout, a linear layer, batch normalization, ReLU, a second dropout, and a final linear layer producing 7-class logits.

Training. Prior to manager training, vision and text logits were pre-extracted and stored to disk for all MELD splits, decoupling manager training from the computationally expensive expert inference and substantially accelerating the hyperparameter search and training loop.

For the vision modality, face crops were first extracted from each MELD video using the OpenCV ResNet-10 SSD face detector [43], sampling every fifth frame and retaining only the highest-confidence detection per sampled frame above a confidence threshold of 0.5. Each detected face crop was converted to grayscale, resized to 48×48 pixels, and normalized

to the $(-1, 1)$ range to match the FER2013+ preprocessing pipeline exactly. Utterances where no face was detected in any sampled frame were discarded. The finetuned vision expert was then run in inference mode over all extracted face crops, producing a sequence of per-frame logits of shape $[T, 7]$ per utterance, where T is the number of detected frames. These logit sequences were saved to disk as pre-extracted features.

For the text modality, the original pretrained text expert was run over all MELD utterances, producing a single utterance-level logit vector of shape $[7]$ per sample. The same pre-extraction procedure was applied to train, development, and test splits for both modalities.

Three inference modes. Batch mode processes complete logit sequences jointly and is used during training and evaluation. Streaming mode updates the GRU hidden state frame by frame, providing a continuously updated emotion estimate from the live camera feed. Anchored mode combines the accumulated streaming hidden state with text logits at message-send time, producing a final prediction that integrates temporal visual context with linguistic content. This design reflects a realistic interaction scenario where visual information is available continuously but text arrives only when the user sends a message.

H. Hyperparameter Optimization with APSO

Both the vision and text experts were independently optimized using Accelerated Particle Swarm Optimization (APSO).

The vision expert APSO search covered three parameters: learning rate, dropout rate, and hidden layer size. The text expert APSO search covered five parameters: learning rate for the BiLSTM head, learning rate for the RoBERTa encoder layers, dropout rate, hidden dimension, and pooling mode index. The two experts were optimized independently, consistent with the decoupled design philosophy of the architecture — each expert is self-contained and can be retrained or replaced without affecting the other. APSO was also applied for optimizing the hyperparameters of the manager network, covering four parameters: learning rate, dropout rate, hidden layer size and number of GRU layers.

The APSO search was configured independently for each component reflecting their different computational costs. For the vision and text experts, the search ran for 10 iterations with a swarm of 10 particles and 5 proxy epochs per fitness evaluation. For the manager network, whose training is substantially lighter — operating on pre-extracted logits rather than raw features — the search budget was increased to 15 iterations and 15 particles with 8 proxy epochs per evaluation, allowing a more thorough exploration of the hyperparameter space at comparable computational cost. In all cases the fitness function was macro F1 on the development split, and the seed was fixed to 42 for the APSO and full training.

I. Knowledge-Based System for Emotion-Aware Dialogue

The emotion recognition pipeline is integrated with a locally deployed conversational agent powered by Ollama, enabling

emotion-aware dialogue without transmitting user data to external services. Rather than using the manager network’s fused prediction directly, the conversational integration employs a symbolic knowledge-based system (KBS) implemented in SWI-Prolog that reasons over the independent predictions and confidence scores of both experts.

Motivation. The KBS approach is motivated by the interpretability limitations of learned fusion and the metrics of the manager. The manager network produces a fused prediction whose reasoning process is opaque — it is impossible to inspect why a particular modality was weighted more heavily in a given prediction. The KBS makes the fusion logic explicit and formally defined, enabling transparent reasoning about modality agreement, conflict, and the psychologically motivated case of emotional masking. This interpretability is particularly valuable in a conversational agent context where understanding why the system responded empathetically to a specific input is important for trust and debugging.

Knowledge base structure. The Prolog knowledge base encodes domain knowledge across three levels.

At the emotion level, emotions are classified by polarity (positive: happiness, surprise; negative: sadness, anger, fear, disgust; neutral: neutral) and intensity (high intensity: anger, fear, disgust; all remaining emotions are treated as low intensity by default). At the modality reasoning level, rules define seven inter-modality relationships. Full agreement occurs when both experts predict the same emotion. Partial agreement occurs when both predict the same polarity but different specific emotions. Full conflict occurs when predictions have opposite polarity. Neutral override occurs when one expert predicts neutral and the other predicts a specific emotion, with the system trusting the non-neutral prediction. Irony or sarcasm is flagged when the vision expert predicts any negative emotion while the text expert predicts a positive emotion, interpreted as a signal that the verbal content contradicts the expressed affect. Emotional masking is detected in two distinct configurations grounded in psychological literature on emotion suppression and social display rules: when the vision expert predicts sadness and the text expert predicts a positive emotion, and when the vision expert predicts any negative emotion and the text expert predicts neutral. Finally, an uncertain case is triggered when both confidence scores fall below a minimum threshold, indicating insufficient signal from either modality. At the response strategy level, rules map each combination of dominant emotion, confidence level, and inter-modality case to one of six response strategies: empathetic priority for high-intensity negative emotions with high confidence, acknowledge and adapt for negative emotions with high confidence, reinforce positive for positive emotions, irony aware for any irony case regardless of confidence, gentle acknowledgement for ambiguous or medium-confidence cases, and neutral supportive for low-confidence or neutral predictions.

Confidence-based arbitration. When modalities conflict, the KBS trusts the higher-confidence prediction rather than defaulting to either modality unconditionally. In the emotional masking case specifically, vision is privileged over text regardless of confidence, reflecting the observation that facial

expressions are more difficult to consciously suppress than word choice.

Python-Prolog integration. The knowledge base is queried from Python via the pyswip library, which interfaces with SWI-Prolog at runtime. The reasoning output — dominant emotion, confidence level, and response strategy — is translated into a natural language emotional context string injected into the Ollama system prompt. The language model is instructed to adapt its tone and response style to the user’s emotional state without explicitly disclosing that emotion analysis is being performed, preserving conversational naturalness. The complete pipeline runs locally with no external API calls, making it suitable for privacy-sensitive deployments.

IV. EVALUATION

A. Experimental Setup

All experiments were conducted on a single NVIDIA GPU using PyTorch 2.11.0+cu126 [44]. The vision expert and text expert were trained on FER2013+ and GoEmotions respectively before finetuning on MELD, with batch sizes of 128 and 64 respectively. The manager network was trained on pre-extracted logits from the finetuned vision expert and the original text expert with a batch size of 128. Hyperparameters for all three components were determined through Accelerated Particle Swarm Optimization (APSO) as described in Section III, with macro F1 on the development splits used as the fitness function throughout. The optimal hyperparameters found by APSO are reported in Table II. The knowledge-based system was implemented in SWI-Prolog and integrated via the pyswip Python interface. The conversational backbone is a locally deployed Ollama instance running Gemma 3 1B [45], accessed via the Ollama Python API without any external network calls.

TABLE II
OPTIMAL HYPERPARAMETERS FOUND BY APSO FOR EACH COMPONENT.
LEARNING RATES ARE RECOVERED FROM LOG SCALE ($lr = 10^x$).

Parameter	Vision	Text	Manager
Learning rate	2.04×10^{-3}	4.48×10^{-4}	6.62×10^{-4}
LR RoBERTa	—	8.87×10^{-5}	—
Dropout	0.34	0.37	0.15
Hidden units	401	407	134
GRU layers	—	—	1
Pooling mode	—	max	—

It is important to note that during text expert training the pooling mode used was mean, after some experiments gave the best result, while the final evaluation was performed using max pooling as returned by the APSO search. This discrepancy between proxy training and final evaluation pooling modes may have introduced a slight inconsistency in the optimization process and represents a limitation of the current hyperparameter search implementation for the text expert.

B. Evaluation Metrics

Emotion recognition on each dataset is evaluated using four metrics. Accuracy measures the proportion of correctly classified utterances across all classes but is a poor indicator

of performance on imbalanced datasets — a model predicting neutral for every utterance would achieve approximately 47% accuracy on MELD’s training split while learning nothing useful. Weighted F1 computes the F1 score per class and averages it weighted by class frequency, rewarding performance on majority classes. Macro F1 computes the F1 score per class and averages them with equal weight regardless of class frequency, treating all seven emotion categories as equally important. Given all datasets severe class imbalance, macro F1 is adopted as the primary evaluation metric throughout this work, as it most faithfully reflects the system’s ability to recognize the full spectrum of emotional states rather than simply predicting the majority class.

C. Experts Results

Table III reports the per-class performance of the vision expert on the FER2013+ test split.

TABLE III
VISION EXPERT (VISIONNET) EVALUATION ON FER2013+ TEST SPLIT.

Emotion	Precision	Recall	F1	Support
Neutral	0.87	0.81	0.84	1,262
Happiness	0.92	0.91	0.92	928
Surprise	0.83	0.86	0.85	444
Sadness	0.62	0.73	0.67	444
Anger	0.77	0.83	0.80	325
Disgust	0.69	0.48	0.56	23
Fear	0.67	0.56	0.61	93
Macro avg	0.77	0.74	0.75	3,519
Weighted avg	0.83	0.83	0.83	3,519
Accuracy	0.83		3,519	

The vision expert achieves a macro F1 of 0.75 and accuracy of 0.83 on the FER2013+ test split, demonstrating strong performance on its training domain. Happiness and surprise obtain the highest F1 scores of 0.92 and 0.85 respectively, benefiting from their distinctive and visually unambiguous facial muscle patterns. Sadness, anger and fear achieve moderate performance in the 0.61–0.67 range, while disgust remains the most challenging class at 0.56 F1 — a consequence of its extremely limited support of only 23 test samples, representing under 1% of the test split.

The vision expert achieves a macro F1 of 0.75 and accuracy of 0.83 on the FER2013+ test split, which represents a competitive performance when compared to specialized benchmarks such as the CNN-based architecture proposed by Patel [46], which reports an accuracy of 0.88 and a macro F1 of 0.87 on the same dataset. Consistent with the state-of-the-art, both systems demonstrate peak performance in the Happiness category, with the current vision expert achieving an F1-score of 0.92 compared to Patel’s 0.95. Notably, this work shows superior performance in identifying Surprise, reaching an F1-score of 0.85, effectively outperforming Patel’s reported 0.83.

However, a significant performance gap remains in minority and subtle classes. While Patel achieves high robustness in categories such as Sadness (0.87) and Fear (0.81), the current vision expert attains 0.67 and 0.61, respectively. The most pronounced discrepancy is found in the Disgust class, where this work obtains an F1 of 0.56 against Patel’s 0.88.

These results establish the upper bound of vision expert performance, which degrades significantly upon transfer to MELD due to the domain shift between controlled FER2013+ images and in-the-wild TV series footage, as discussed in Section IV-D.

Table IV reports the per-class performance of the text expert on the GoEmotions test split.

TABLE IV
TEXT EXPERT (ROBERTA-BiLSTM) EVALUATION ON GOEMOTIONS TEST SPLIT.

Emotion	Precision	Recall	F1	Support
Neutral	0.66	0.51	0.57	1,606
Happiness	0.81	0.81	0.81	2,054
Surprise	0.50	0.69	0.58	573
Sadness	0.58	0.50	0.53	317
Anger	0.49	0.62	0.55	703
Disgust	0.46	0.33	0.39	84
Fear	0.64	0.68	0.66	90
Macro avg	0.59	0.59	0.58	5,427
Weighted avg	0.67	0.66	0.66	5,427
Accuracy	0.66		5,427	

Table IV reports the text expert evaluation on the GoEmotions test split. The model achieves a macro F1 of 0.58 and weighted F1 of 0.66, reflecting the expected performance gap between majority and minority classes. Happiness obtains the highest F1 of 0.81, benefiting from its dominant representation in the training distribution (39% of samples). Minority classes show notably lower performance — disgust reaches only 0.39 F1 despite asymmetric augmentation, reflecting the fundamental difficulty of learning from 580 training samples. Fear performs surprisingly well at 0.66 despite its similarly limited support, likely due to its distinctive lexical patterns. Surprise and anger achieve moderate F1 scores of 0.58 and 0.55 respectively. These results are consistent with the macro F1 of 0.64 reported by [47] for a transformer-based model (BERT) on the same Ekman-mapped GoEmotions benchmark. These results establish a strong baseline for the text modality, which is subsequently transferred zero-shot to the MELD domain without finetuning — a decision validated by the degradation observed when finetuning was attempted, as is discussed in Section IV-D.

Table V reports the performance of both unimodal experts on the MELD test split, comparing zero-shot transfer performance (before finetuning) against the final finetuned models.

TABLE V
PER-CLASS F1 SCORE COMPARISON BEFORE AND AFTER FINETUNING ON MELD TEST SPLIT FOR BOTH UNIMODAL EXPERTS. ↑ INDICATES IMPROVEMENT, ↓ INDICATES DEGRADATION.

Emotion	Vision Expert		Text Expert	
	Pre-FT	Post-FT	Pre-FT	Post-FT
Neutral	0.58	0.45 ↓	0.54	0.40 ↓
Happiness	0.30	0.29 ↓	0.42	0.43 ↑
Surprise	0.00	0.17 ↑	0.44	0.42 ↓
Sadness	0.10	0.19 ↑	0.29	0.31 ↑
Anger	0.05	0.07 ↑	0.29	0.25 ↓
Disgust	0.00	0.00 —	0.00	0.03 ↑
Fear	0.00	0.03 ↑	0.12	0.12 —
Macro F1	0.15	0.17 ↑	0.30	0.28 ↓
Weighted F1	0.34	0.30 ↓	0.43	0.37 ↓
Accuracy	0.41	0.29 ↓	0.44	0.36 ↓

For the vision expert, finetuning produces a marginal macro F1 improvement from 0.15 to 0.17, driven by gains in previously zero-performance classes — surprise improves from 0.00 to 0.17, sadness from 0.10 to 0.19, and fear from 0.00 to 0.03. This improvement comes at the cost of majority class performance, with neutral F1 dropping from 0.58 to 0.45 and overall accuracy decreasing from 0.41 to 0.29. This tradeoff is the intended effect of the `WeightedRandomSampler` and macro F1 optimization strategy — the model redistributes predictive capacity from the dominant neutral class toward underrepresented emotions. Since macro F1 is the primary evaluation metric in this work, the finetuned vision expert is selected for all subsequent experiments.

For the text expert the opposite pattern emerges — finetuning degrades performance across most metrics, with macro F1 dropping from 0.30 to 0.28, accuracy from 0.44 to 0.36, and weighted F1 from 0.43 to 0.37. Only happiness and sadness show marginal gains. The pretrained RoBERTa-BiLSTM model trained on GoEmotions generalizes more robustly to MELD than the finetuned version, likely due to the combination of MELD’s severe class imbalance and the domain shift from Reddit comments to TV series dialogue causing the model to overfit to the neutral-dominated training distribution. The original text expert is therefore used in all subsequent experiments.

D. Manager Results

Table VI reports the per-class performance of the manager network on the MELD test split.

TABLE VI
MANAGER NETWORK EVALUATION ON MELD TEST SPLIT.

Emotion	Precision	Recall	F1	Support
Neutral	0.68	0.26	0.38	448
Happiness	0.40	0.58	0.47	165
Surprise	0.31	0.59	0.41	112
Sadness	0.19	0.36	0.25	66
Anger	0.22	0.32	0.26	107
Disgust	0.06	0.04	0.05	27
Fear	0.03	0.04	0.04	26
Macro avg	0.27	0.31	0.26	951
Weighted avg	0.47	0.36	0.36	951
Accuracy	0.36			951

The manager network achieves a macro F1 of 0.26 and an accuracy of 0.36 on the MELD test split, reflecting the substantial challenges posed by MELD’s domain characteristics and severe class imbalance. a random classifier respecting uniform class priors would achieve an accuracy of approximately 0.14, placing the manager meaningfully above random guessing. These results are consistent with the difficulty of the task, although they sit below current state-of-the-art multimodal systems such as MultiEMO [48], which achieves a weighted F1 of 0.67 and an accuracy of 0.67 on the same benchmark.

The performance gap is primarily driven by MultiEMO’s use of bidirectional multi-head cross-attention and a Sample-Weighted Focal Contrastive (SWFC) loss, which specifically targets the classification of minority and semantically similar emotions. While the manager network struggles with minority

classes—obtaining near-zero F1 scores for Disgust (0.05) and Fear (0.04)—MultiEMO significantly elevates these metrics to 0.37 and 0.30 respectively.

Furthermore, the manager’s performance is constrained by its lack of conversational context. While MultiEMO utilizes DialogueRNN and RoBERTa to model speaker dependencies and temporal flow, the manager network treats each utterance as an isolated event. This is particularly evident in the Neutral class: while MultiEMO achieves a robust F1 of 0.78, the manager network reaches only 0.38, suffering from a poor recall of 0.26 due to the `WeightedRandomSampler` pushing the model away from the majority class.

Finally, it is important to note the domain gap. MultiEMO is trained end-to-end on MELD, whereas the manager operates on experts pre-trained on external datasets (FER2013+ for vision and GoEmotions for text). MultiEMO further optimizes the visual modality by using MTCNN to isolate facial regions, reducing background noise that the manager’s vision expert likely encounters in MELD’s in-the-wild footage. These architectural choices—context modeling, correlation-aware fusion, and specialized loss—define the performance ceiling that end-to-end systems like MultiEMO can reach compared to modular expert-based fusion.

E. Knowledge-Based System Qualitative Analysis

The results demonstrate both the capabilities and current limitations of the KBS-LLM integration. The high-intensity agreement case (fear + fear) produces the most notably empathetic response — the LLM slows its pace, explicitly validates the anxiety, offers grounding techniques, and invites the user to share more without pressure, consistent with the *empathetic_priority* strategy. The neutral override case produces a warm and emotionally present response to a message about loss, correctly overriding the neutral face signal in favour of the sadness detected in the text. The irony case produces a response that acknowledges the difficult nature of Mondays with a knowing tone and gently redirects toward constructive action, correctly reflecting the *irony_aware* strategy rather than responding to the literal positive content of the message.

The masking case presents a mixed result. Despite the KBS correctly classifying the interaction as masking and assigning the *gentle_acknowledgement* strategy, the LLM opens cheerfully with “That’s wonderful to hear!” — responding to the surface content of the message rather than the underlying sadness signal. However, the response immediately pivots toward an offer of support and gently invites the user to share if they feel comfortable, which partially reflects the intended strategy. This suggests that while the KBS reasoning is correctly propagated to the language model, the model’s tendency to respond to literal word meaning before consulting the emotional context limits full alignment. Stronger prompt engineering or a more instruction-following model would likely improve this behaviour, and represents a direction for future work.

V. CONCLUSION

This work presented a multimodal emotion recognition system combining facial expression and text analysis through

TABLE VII
REPRESENTATIVE KBS FUSION CASES WITH CORRESPONDING LLM RESPONSES. VISION AND TEXT PREDICTIONS SHOWN WITH SOFTMAX CONFIDENCE. RESPONSES TRUNCATED FOR SPACE.

Message	Vision	Text	Case	Strategy	LLM Response (excerpt)
"I'm fine, don't worry about me."	sadness (72%)	neutral (65%)	masking	gentle_ack.	"That's wonderful to hear! Is there anything I can do to help you feel a little better, or anything on your mind you'd like to share?"
"Oh great, another Monday. Just what I needed."	anger (60%)	happiness (55%)	irony	irony_aware	"Monday, huh? It's tough, isn't it? How about you do something to make it a little better, even just for a moment?"
"I have an exam tomorrow and I can't stop thinking about it."	fear (85%)	fear (80%)	agreement	empathetic_priority	"I hear you. It sounds like a really significant amount of pressure [...] Take a deep breath. It's okay to feel this way."
"My dog passed away last week."	neutral (75%)	sadness (60%)	neutral_override	gentle_ack.	"I'm so incredibly sorry for your loss. Losing a dog is like losing a piece of your heart [...] I'm here to listen if you need to talk."

two independent fusion strategies for emotion-aware conversational interaction. A VGG-style vision expert trained on FER2013+ and a RoBERTa-BiLSTM text expert trained on GoEmotions were optimized through Accelerated Particle Swarm Optimization and subsequently evaluated on the MELD benchmark, achieving macro F1 scores of 0.75 and 0.58 on their respective training domains. A GRU-based manager network trained on pre-extracted expert logits achieved a macro F1 of 0.26 on MELD, operating without audio under a more constrained input configuration than comparable systems in the literature.

Accelerated Particle Swarm Optimization proved an effective metaheuristic strategy for hyperparameter search across all three learned components. The most notable result was achieved on the vision expert, which reached a macro F1 of 0.75 on FER2013+ consistent with CNN-based architectures reported in the literature [46]. The text expert achieved competitive results on GoEmotions, though the pooling mode hyperparameter required manual adjustment beyond the APSO search to reach optimal performance, suggesting that discrete architectural choices benefit from additional targeted exploration beyond standard swarm-based optimization.

The knowledge-based system represents the primary architectural contribution of this work — a rule-based expert system implemented in SWI-Prolog that explicitly models psychological phenomena including emotional masking, irony detection, and neutral override, deriving a response strategy that is injected into a locally deployed large language model. Qualitative evaluation demonstrated that the KBS produces meaningfully differentiated conversational responses across inter-modality conflict cases that purely statistical fusion approaches do not explicitly model, including sarcasm detection and emotional suppression recognition.

Several limitations constrain the current system. MELD's severe class imbalance — neutral representing 47% of training samples — imposes a hard ceiling on minority class performance regardless of architecture. The domain gap between FER2013+ controlled images and MELD's in-the-wild video footage limits vision expert generalization. The absence of an audio modality, discarded due to cross-corpus domain shift from RAVDESS to MELD, removes a signal that state-of-the-art systems rely on heavily. Finally, the alignment between

KBS-derived strategies and LLM response behaviour depends on prompt engineering quality, as evidenced by the masking case where the model responded to surface content rather than the underlying emotional signal.

The modular design of the proposed system — where each expert is trained independently and fusion operates on pre-extracted logits — offers practical advantages in maintainability, missing modality robustness, and component replaceability. However, this modularity comes at a measurable performance cost relative to end-to-end systems such as MultiEMO, which optimize all components jointly on the target domain. This trade-off motivated the development of the KBS as an independent alternative fusion strategy — one that foregoes learned optimization in favour of explicit psychological reasoning, capturing inter-modality phenomena such as masking and irony that purely statistical approaches do not model. Nevertheless, future work should explore whether partial end-to-end training — finetuning the full pipeline jointly on MELD while retaining the modular inference architecture — can narrow this gap without sacrificing the system's practical deployment advantages.

VI. FUTURE WORK

Future work should address these limitations through several directions. End-to-end training of the full multimodal pipeline on MELD, rather than the current modular logit-based approach, would reduce the domain gap penalty. Audio reintegration using a speech encoder pretrained on conversational data rather than acted corpora such as RAVDESS would restore the third modality. Automatic prompt optimization or finetuning of the language model on strategy-conditioned response pairs would improve KBS-LLM alignment. Finally, extending the KBS rule base with additional psychological reasoning patterns — including cultural variation in emotional expression and speaker-specific baseline modeling — would increase the system's robustness in naturalistic interaction scenarios.

REFERENCES

- [1] M. J. Dileep Kumar, M. Sukesh Rao, and K. C. Narendra, "Multimodal emotion recognition: A comprehensive survey of datasets, methods, and applications," *IEEE Access*, vol. 13, pp. 201 067–201 097, 2025.

- [2] Y. Shou, T. Meng, W. Ai, F. Fu, N. Yin, and K. Li, "A comprehensive survey on multi-modal conversational emotion recognition with deep learning," *ACM Trans. Inf. Syst.*, vol. 44, no. 2, Jan. 2026. [Online]. Available: <https://doi.org/10.1145/3786343>
- [3] P. Ekman, "An argument for basic emotions," *Cognition & Emotion*, vol. 6, pp. 169–200, 1992. [Online]. Available: <https://api.semanticscholar.org/CorpusID:11771973>
- [4] Y. Wang, W. Song, W. Tao, A. Liotta, D. Yang, X. Li, S. Gao, Y. Sun, W. Ge, W. Zhang, and W. Zhang, "A systematic review on affective computing: Emotion models, databases, and recent advances," 03 2022.
- [5] J. Russell, "A circumplex model of affect," *Journal of Personality and Social Psychology*, vol. 39, pp. 1161–1178, 12 1980.
- [6] R. Plutchik, "The Nature of Emotions," *American Scientist*, vol. 89, no. 4, p. 344, Jul. 2001.
- [7] N. Dalal and B. Triggs, "Histograms of oriented gradients for human detection," vol. 1, 07 2005, pp. 886–893.
- [8] T. Ojala, M. Pietikäinen, and T. Mäenpää, "Multiresolution gray-scale and rotation invariant texture classification with local binary patterns," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 7, p. 971–987, Jul. 2002. [Online]. Available: <https://doi.org/10.1109/TPAMI.2002.1017623>
- [9] J. Lien, T. Kanade, J. Cohn, and C. Li, "Automated facial expression recognition based on face action units," 01 1998, pp. 390–395.
- [10] S. Sudhanshu and A. Khalid, "A custom convolutional neural network for facial emotion recognition on fer2013: Evaluation with 9-fold cross-validation," in *Proceedings of the 2025 International Conference on Artificial Intelligence and Its Applications*, ser. icARTi '25. New York, NY, USA: Association for Computing Machinery, 2025. [Online]. Available: <https://doi.org/10.1145/3774791.3774797>
- [11] J. Haddad, O. Lezoray, and P. Hamel, "3d-cnn for facial emotion recognition in videos," in *Advances in Visual Computing: 15th International Symposium, ISVC 2020, San Diego, CA, USA, October 5–7, 2020, Proceedings, Part II*. Berlin, Heidelberg: Springer-Verlag, 2020, p. 298–309. [Online]. Available: https://doi.org/10.1007/978-3-030-64559-5_23
- [12] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," *ArXiv*, vol. abs/2010.11929, 2020. [Online]. Available: <https://api.semanticscholar.org/CorpusID:225039882>
- [13] D. Bhagat, A. Vakil, R. K. Gupta, and A. Kumar, "Facial emotion recognition (fer) using convolutional neural network (cnn)," *Procedia Computer Science*, vol. 235, pp. 2079–2089, 2024, international Conference on Machine Learning and Data Engineering (ICMLDE 2023). [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1877050924008731>
- [14] C. Strapparava and R. Mihalcea, "Learning to identify emotions in text," in *Proceedings of the 2008 ACM Symposium on Applied Computing*, ser. SAC '08. New York, NY, USA: Association for Computing Machinery, 2008, p. 1556–1560. [Online]. Available: <https://doi.org/10.1145/1363686.1364052>
- [15] S. Mohammad and P. Turney, "Crowdsourcing a word-emotion association lexicon," *Computational Intelligence*, vol. 29, 08 2013.
- [16] A. Valitutti, C. Strapparava, and O. Stock, "Developing affective lexical resources," *Psychology Journal*, vol. 2, pp. 61–83, 01 2004.
- [17] M. M. Bradley and P. J. Lang, "Affective norms for english words (anew): Instruction manual and affective ratings," 1999. [Online]. Available: <https://api.semanticscholar.org/CorpusID:145474983>
- [18] E. Cambria, D. Das, S. Bandyopadhyay, and A. Feraco, *Affective Computing and Sentiment Analysis*. Cham: Springer International Publishing, 2017, pp. 1–10.
- [19] T. Joachims, "Text categorization with support vector machines: Learning with many relevant features," in *Machine Learning: ECML-98*, C. Nédellec and C. Rouveil, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 1998, pp. 137–142.
- [20] A. McCallum and K. Nigam, "A comparison of event models for naive bayes text classification," in *AAAI Conference on Artificial Intelligence*, 1998. [Online]. Available: <https://api.semanticscholar.org/CorpusID:7311285>
- [21] Y. Kim, "Convolutional neural networks for sentence classification," in *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, A. Moschitti, B. Pang, and W. Daelemans, Eds. Doha, Qatar: Association for Computational Linguistics, Oct. 2014, pp. 1746–1751. [Online]. Available: <https://aclanthology.org/D14-1181/>
- [22] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, J. Burstein, C. Doran, and T. Solorio, Eds. Minneapolis, Minnesota: Association for Computational Linguistics, Jun. 2019, pp. 4171–4186. [Online]. Available: <https://aclanthology.org/N19-1423/>
- [23] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, "Roberta: A robustly optimized bert pretraining approach," 07 2019.
- [24] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, "Language models are unsupervised multitask learners," 2019. [Online]. Available: <https://api.semanticscholar.org/CorpusID:160025533>
- [25] R. S. Sitoula, M. Pramanik, and R. Panigrahi, "Fine-grained classification for emotion detection using advanced neural models and GoEmotions dataset," *Journal of Soft Computing and Data Mining*, vol. 5, no. 2, pp. 62–71, 2024.
- [26] A. Harutyunyan and S. Kumar, "Fine-grained emotion detection on goemotions: Experimental comparison of classical machine learning, bilstm, and transformer models," 01 2026.
- [27] N. Penumajji, "Deep learning for speech emotion recognition: A cnn approach utilizing mel spectrograms," 03 2025.
- [28] Y. Gong, Y.-A. Chung, and J. Glass, "AST: Audio Spectrogram Transformer," in *Interspeech 2021*, 2021, pp. 571–575.
- [29] Y.-H. H. Tsai, S. Bai, P. P. Liang, J. Z. Kolter, L.-P. Morency, and R. Salakhutdinov, "Multimodal transformer for unaligned multimodal language sequences," in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, A. Korhonen, D. Traum, and L. Márquez, Eds. Florence, Italy: Association for Computational Linguistics, Jul. 2019, pp. 6558–6569. [Online]. Available: <https://aclanthology.org/P19-1656/>
- [30] N. Majumder, S. Poria, D. Hazarika, R. Mihalcea, A. Gelbukh, and E. Cambria, "Dialogueernn: An attentive rnn for emotion detection in conversations," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, pp. 6818–6825, 07 2019.
- [31] Dumitru, I. Goodfellow, W. Cukierski, and Y. Bengio, "Challenges in representation learning: Facial expression recognition challenge," <https://kaggle.com/competitions/challenges-in-representation-learning-facial-expression-recognition-challenge>, 2013, kaggle.
- [32] E. Barsoum, C. Zhang, C. Canton Ferrer, and Z. Zhang, "Training deep networks for facial expression recognition with crowd-sourced label distribution," in *ACM International Conference on Multimodal Interaction (ICMI)*, 2016.
- [33] D. Demszky, D. Movshovitz-Attias, J. Ko, A. Cowen, G. Nemade, and S. Ravi, "GoEmotions: A dataset of fine-grained emotions," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, D. Jurafsky, J. Chai, N. Schluter, and J. Tetreault, Eds. Online: Association for Computational Linguistics, Jul. 2020, pp. 4040–4054. [Online]. Available: <https://aclanthology.org/2020.acl-main.372/>
- [34] S. R. Livingstone and F. A. Russo, "The ryerson audio-visual database of emotional speech and song (ravdess): A dynamic, multimodal set of facial and vocal expressions in north american english," *PLOS ONE*, vol. 13, no. 5, pp. 1–35, 05 2018. [Online]. Available: <https://doi.org/10.1371/journal.pone.0196391>
- [35] C. Busso, M. Bulut, C.-C. Lee, A. Kazemzadeh, E. Mower Provost, S. Kim, J. Chang, S. Lee, and S. Narayanan, "Iemocap: Interactive emotional dyadic motion capture database," *Language Resources and Evaluation*, vol. 42, pp. 335–359, 12 2008.
- [36] S. Poria, D. Hazarika, N. Majumder, G. Naik, E. Cambria, and R. Mihalcea, "MELD: A multimodal multi-party dataset for emotion recognition in conversations," in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, A. Korhonen, D. Traum, and L. Márquez, Eds. Florence, Italy: Association for Computational Linguistics, Jul. 2019, pp. 527–536. [Online]. Available: <https://aclanthology.org/P19-1050/>
- [37] Z. Song, X. Zheng, L. Liu, M. Xu, and X. Huang, "Generating responses with a specific emotion in dialog," in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, A. Korhonen, D. Traum, and L. Márquez, Eds. Florence, Italy: Association for Computational Linguistics, Jul. 2019, pp. 3685–3695. [Online]. Available: <https://aclanthology.org/P19-1359/>
- [38] Z. Lin, A. Madotto, J. Shin, P. Xu, and P. Fung, "MoEL: Mixture of empathetic listeners," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, K. Inui, J. Jiang, V. Ng, and X. Wan, Eds. Hong

- Kong, China: Association for Computational Linguistics, Nov. 2019, pp. 121–132. [Online]. Available: <https://aclanthology.org/D19-1012/>
- [39] M. Ibrahim, N. Hussein, D. Guinovart *et al.*, “Optimizing convolutional neural networks: A comprehensive review of hyperparameter tuning through metaheuristic algorithms,” *Archives of Computational Methods in Engineering*, vol. 32, pp. 5123–5160, 2025.
- [40] A. Gaspar, D. Oliva, E. Cuevas, D. Zaldívar, M. Pérez, and G. Pajares, *Hyperparameter Optimization in a Convolutional Neural Network Using Metaheuristic Algorithms*. Cham: Springer International Publishing, 2021, pp. 37–59. [Online]. Available: https://doi.org/10.1007/978-3-030-70542-8_2
- [41] J. Kennedy and R. C. Eberhart, “Particle swarm optimization-neural networks, 1995. proceedings., ieee international conference on,” 2004. [Online]. Available: <https://api.semanticscholar.org/CorpusID:118262007>
- [42] A.-H. M. Emara, G. Atteia, and J. H. Alkhateeb, “Fine tuning hyperparameters of deep learning models using metaheuristic accelerated particle swarm optimization algorithm,” *IEEE Access*, vol. 13, pp. 134 506–134 518, 2025.
- [43] G. Bradski, *The OpenCV Library*, 11 2000, vol. 25.
- [44] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimesheine, L. Antiga, A. Desmaison, A. Köpf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, and S. Chintala, *PyTorch: an imperative style, high-performance deep learning library*. Red Hook, NY, USA: Curran Associates Inc., 2019.
- [45] Google DeepMind, “Gemma 3 technical report,” *arXiv preprint arXiv:2503.19786*, 2025.
- [46] D. G. Patel, “Facial emotion recognition,” Master’s Project Report, Concordia University of Edmonton, Edmonton, Alberta, August 2023.
- [47] D. Demszky, D. Movshovitz-Attias, J. Ko, A. Cowen, G. Nemade, and S. Ravi, “GoEmotions: A dataset of fine-grained emotions,” in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Online: Association for Computational Linguistics, Jul. 2020, pp. 4040–4054. [Online]. Available: <https://aclanthology.org/2020.acl-main.372>
- [48] T. Shi and S.-L. Huang, “MultiEMO: An attention-based correlation-aware multimodal fusion framework for emotion recognition in conversations,” in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, A. Rogers, J. Boyd-Graber, and N. Okazaki, Eds. Toronto, Canada: Association for Computational Linguistics, Jul. 2023, pp. 14 752–14 766. [Online]. Available: <https://aclanthology.org/2023.acl-long.824/>